



# Graphs Intermediate - Problem Solving

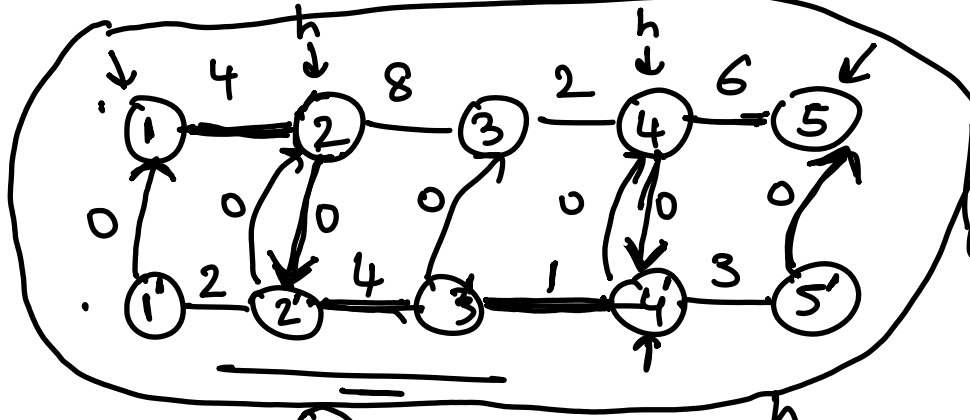
- Anikate Koul

# Problem 1:



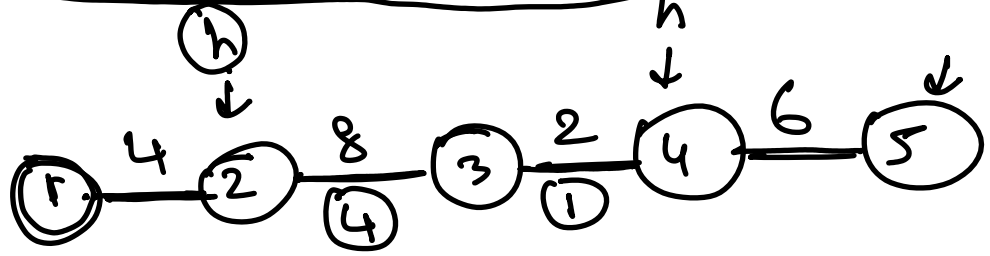
**Rendez-vous de Marian et Robin:**

<https://codeforces.com/contest/2014/problem/E>



$$4 + 0 + 4 + 1 + 6 + 0$$

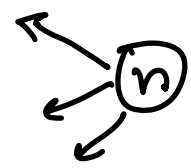
$$= 15 //$$



$$4 + 4 + 1 + 6 = 15 //$$

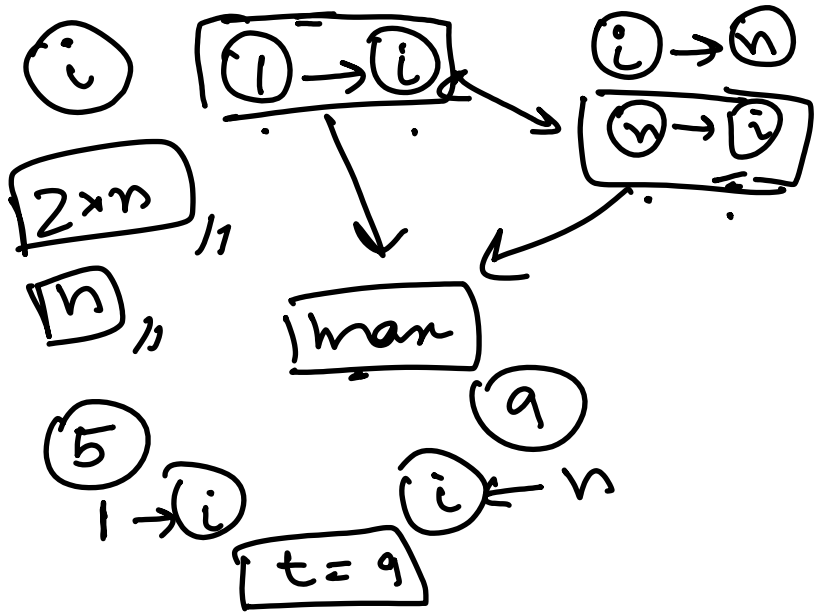
$$4 + 0 + 4 + 1 + 3 + 0$$

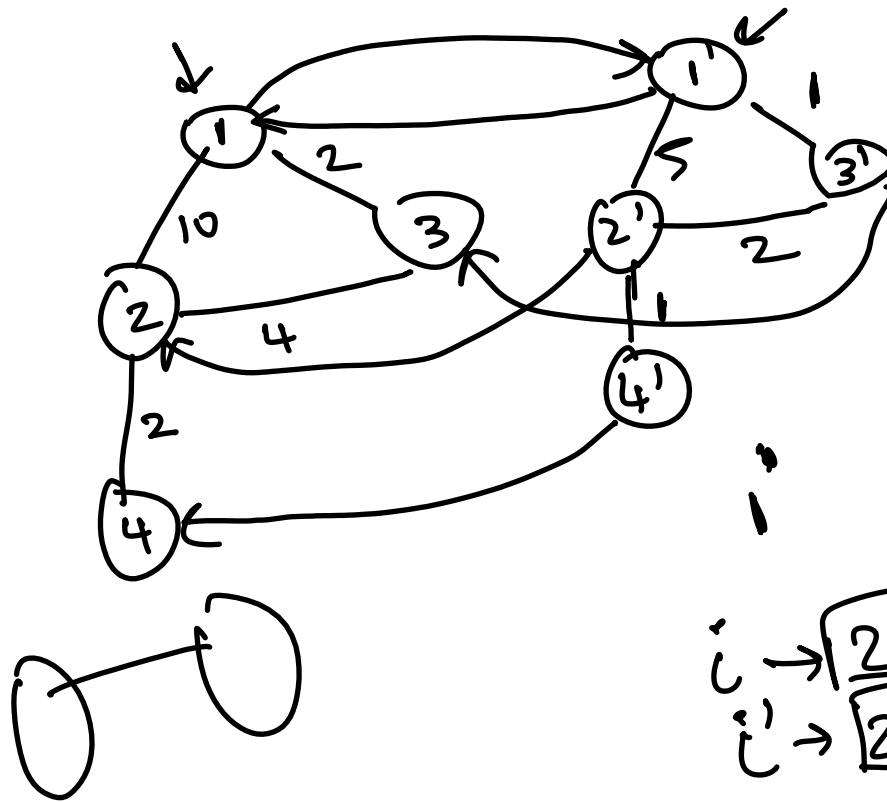
$$= 12 //$$



min(i)

i → 1, n  
1', n'



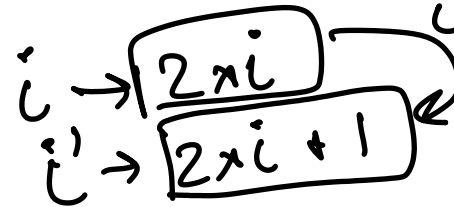


$$i \rightarrow 2 \times i$$

$$i' \rightarrow 2 \times i + 1$$

$$i \rightarrow i$$

$$i' \rightarrow h + i$$



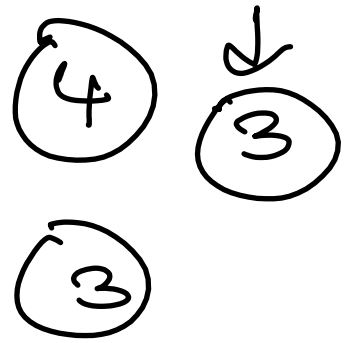
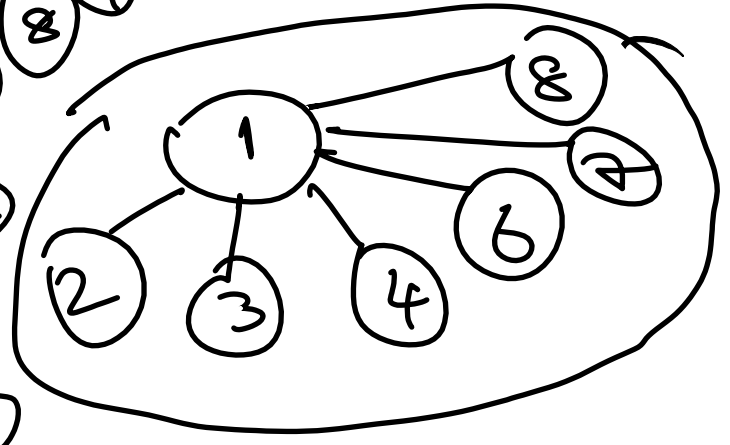
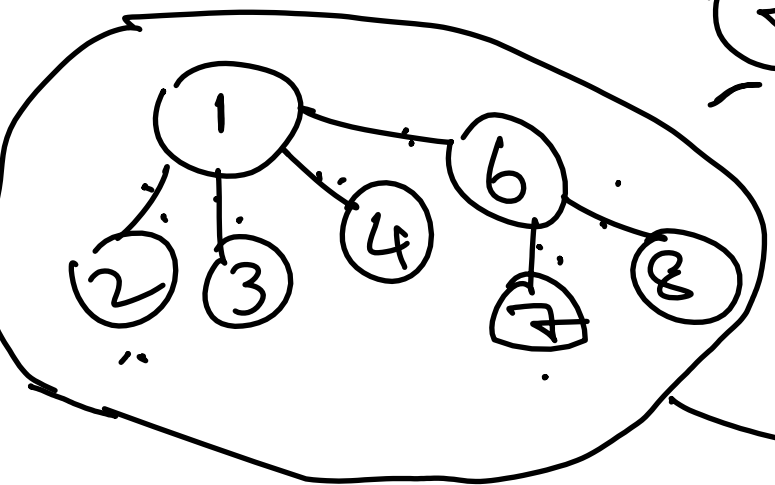
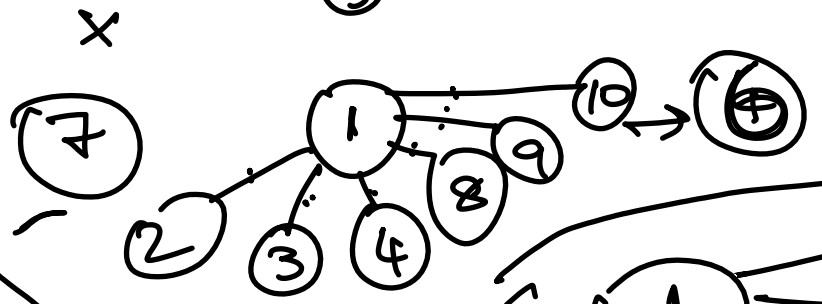
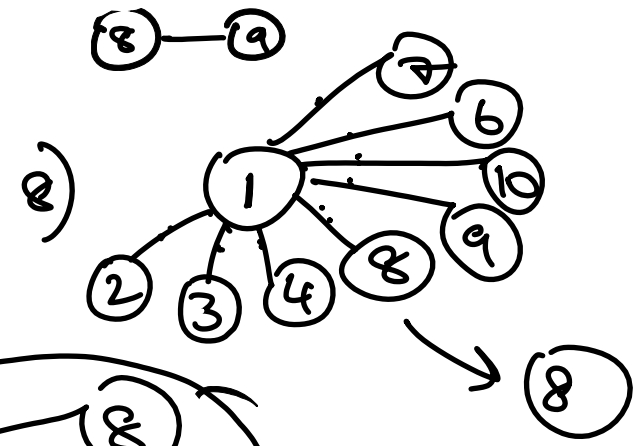
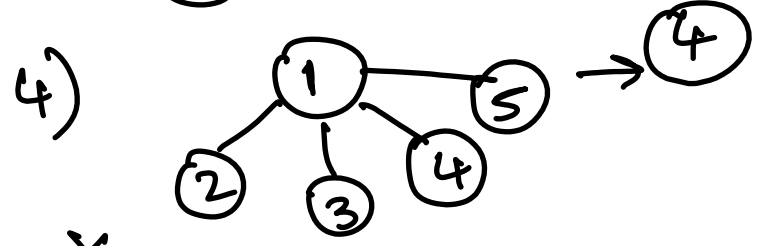
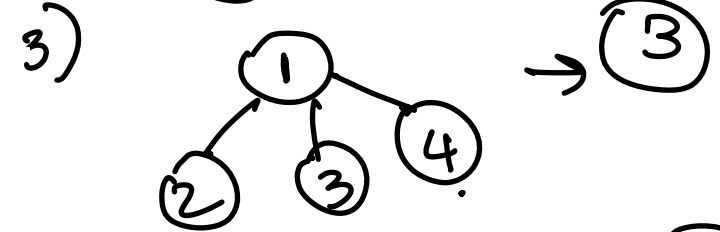
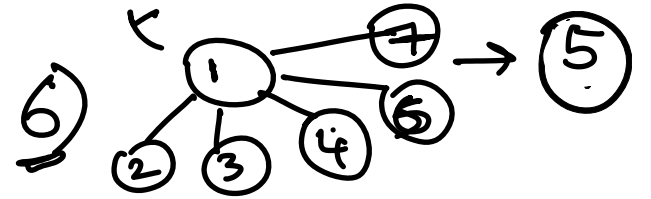
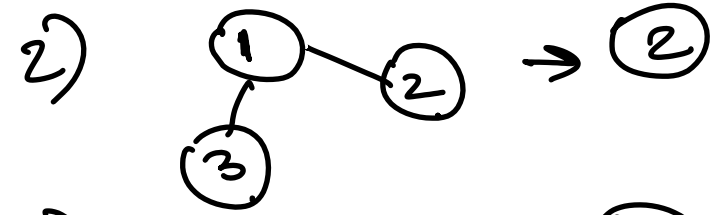
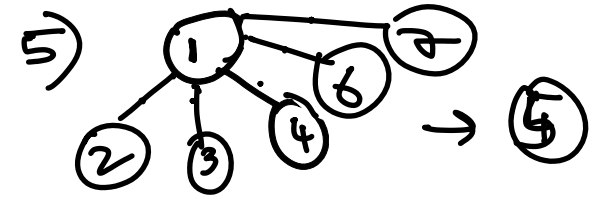
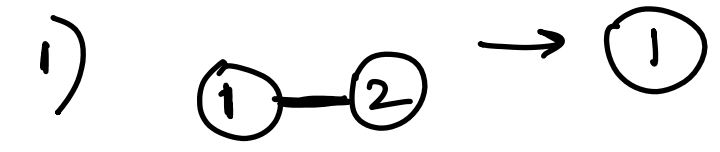
# Problem 2

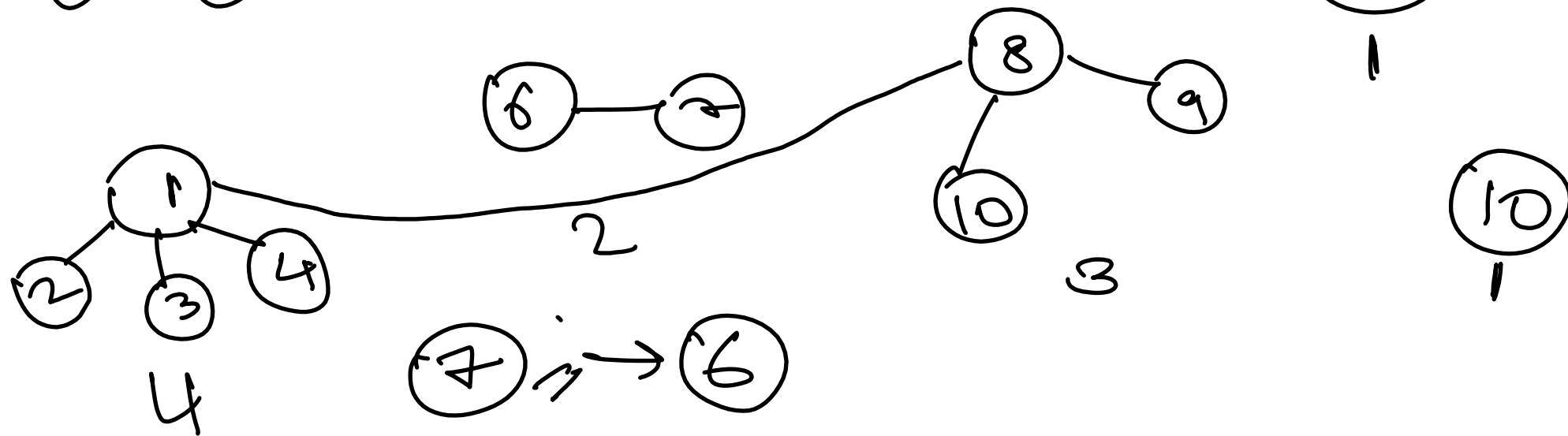
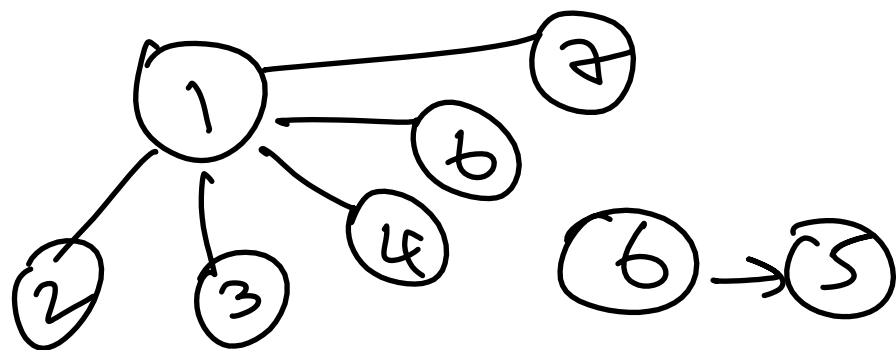
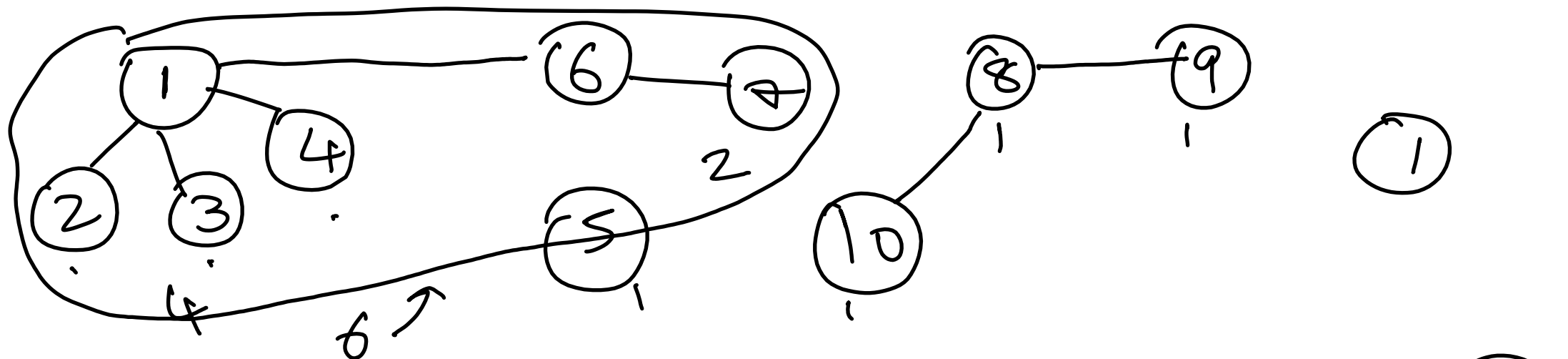


**Social Network:**

<https://codeforces.com/problemset/problem/1609/D>

10 8  
 12  
 23  
 34  
 14  
 67  
 89  
 810  
 14

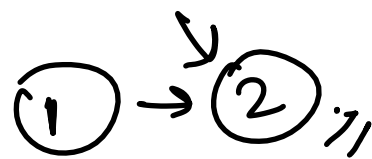




a

b

union(a, b),

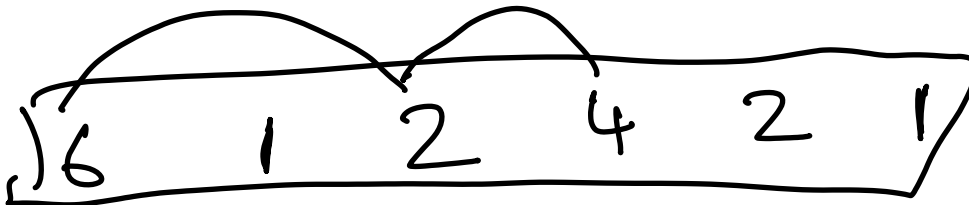


(i)

i+1

(a, b),

(b, c)



(12)

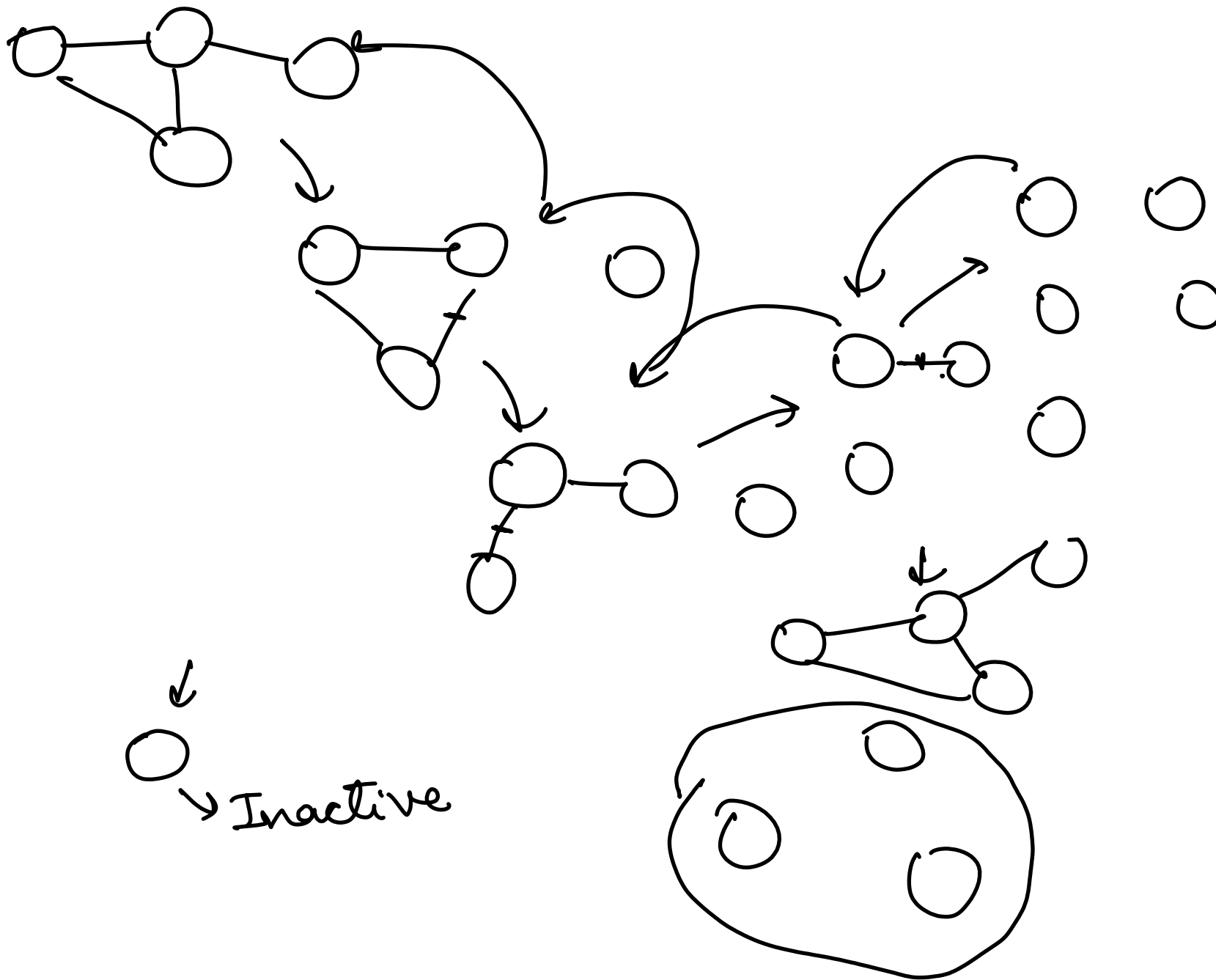
(11)

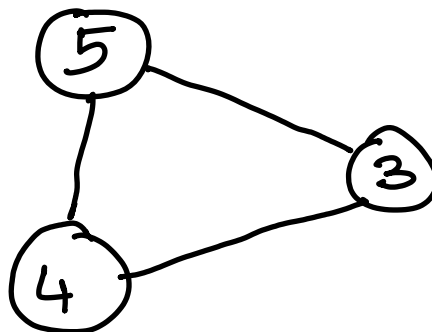
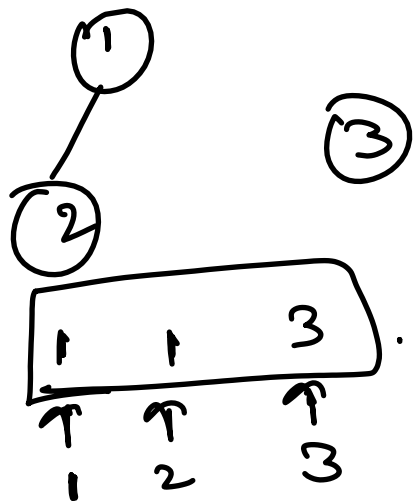
# Problem 3



**Closing the Farm:**

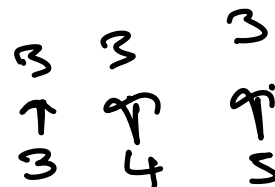
<https://usaco.org/index.php?page=viewproblem2&cpid=646>





$5 \rightarrow 3, 2, 4$       1, 2

$3 \rightarrow 4, 5, 1$



$\{1, \underline{6}\} \rightarrow$

$\{1, 53\}$

